

The Ephemeris

June 2020

Volume 31 Number 02- The Official Publication of the San Jose Astronomical Association



June 2020 - September 2020 Events

Schedule subject to change due to Covid-19 restrictions.

Please check our website for up to date information.

Board & General Meetings

Saturday 6/6, 7/11, 8/8, 8/29,
Board Meetings: 6 -7:30pm
General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 6/7, 7/12, 8/2, 9/13

Solar Sunday (2-4pm)

Sunday 6/7, 7/12, 8/2, 9/13

Intro to the Night Sky Class & Houge Park 1Q In-Town Star Party

Friday 6/26, 7/31, 8/28, 9/25

Houge Park 3Q In-Town Star Party

Friday 6/12, 7/17, 8/14, 9/11

RCDO Starry Nights Star Party

Saturday 6/13, 7/18, 8/15, 9/12

Imaging SIG Mtg

Tuesday 6/16, 7/21, 8/18, 9/15

Astro Imaging Workshops (at Little Uvas OSP)

Saturday 7/18, 8/22

Binocular Star Gazing

Currently on hold

*Please refer to the SJAA Website home page
(www.sjaa.net) and click on any "Upcoming Event" high-
lighted in Blue. It will take you to the SJ-Astronomy
Meetup page ([www.meetup.com/SJ-Astronomy/
events/](http://www.meetup.com/SJ-Astronomy/events/)) where you can find specific event times, loca-
tions, maps and possible cancellation due to weather.*

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Photo by SJAA Member: Glenn Newell

Feb 7-9, 2020

Location: Union City, CA copyright 2020

In Memoriam to Gerry Joyce

Credit: Bruce Braunstein



Gerry Joyce with his telescope



Gerry with some members of his SJAA family

As we are all aware, our club president and friend Gerry Joyce passed away recently. Gerald James Joyce (Gerry) was born on November 2, 1945 and passed away on May 20th 2020. He was 74 years old.

Gerry was born in Long Branch, New Jersey, the youngest of four boys and the son of Norwegian immigrants Frank and Ruth Joyce. After a three-year stint with the Navy, he married his first wife, Nancy, and began a 50-year career in computer technology. Over his lifetime, Gerry worked with both large and small companies, but had a particular “soft spot” for the challenges of startups. He was an innovator in the early days of computers; his natural curiosity, thirst for learning, and tenacity took him through many phases of his career. His entrepreneurial spirit brought him to California in the late 70’s. In 1999 he married Priscilla Wallace, and on November 6, 2019 they celebrated their 20th anniversary in Kauai, Hawaii. Gerry and Priscilla traveled extensively, including a trip to Norway to reacquaint with relatives. In 2004, he started his own company, Viking Systems, to design and install all forms of home automation. He sold Viking in 2016 and retired.

During the last four years, he was able to pursue a life-long interest in astronomy and astrophotography. In March 2020 Gerry was elected President of the San Jose Astronomical Association. Prior to that, he had also served as the Vice-President of SJAA from October 2019 to March 2020. His excitement was contagious – he was always on a mission to interest everyone – from grandchildren to casual acquaintances – in the wonders of the sky. SJAA enriched Gerry’s retirement years immeasurably, and he made many dear friends during his time with the club.

Gerry is survived by his wife, Priscilla; sons Stephen (Briana), Eric (Lisa), and David (Teri); stepson Steve Wallace (Dana), stepdaughter Erin Wallace; grandchildren Vendela, Karsten and Chloe; and step grandchildren Jonas, Cora, Kelsey and Adelyn.

In lieu of flowers, please donate in his name to the Humane Society of Silicon Valley or Second Harvest Food Bank of Silicon Valley. A private memorial service was held on Saturday, June 6th 2020.

Armchair Star Party

Credit: Muditha Kanchana

When the Covid-19 crisis hit, SJAA had to halt all regular star parties. SJAA night- time star parties include In Town Star Party (ITSP) we hold a couple of times a month on Fridays at Houge Park, Monthly Dark Sky "Starry Nights" star party we hold at Rancho Cañada Del Oro monthly, school star parties, and a few irregular events. SJAA leaders discussed how to continue at least a few of the star parties virtually. We decided to move ITSP & Starry nights into a new combined virtual format named "Armchair Star Party". SJAA imaging group offered to help with live view broadcasts from member residences along with the use of existing member images. Moving to an online format also enabled us to focus more on the educational aspect of star parties. Instead of describing the sky, equipment, and other astronomy-related content in a dark setting, presenters get a chance to use visuals to educate participants in an online setting.

The First Armchair Star Party was held on Saturday, May 23rd night. This event was a true collaboration among SJAA Imaging, Starry Nights, & ITSP groups. Many hours of preparation went into the event. It was a challenge to organize multiple speakers, live imaging, and a brand new set of materials. Special mentions go to Rashi for putting together a great set of slides. Likewise to Glenn for his Stellarium customizations and for organizing the live imaging sessions.

The first part of the event was to introduce SJAA, the new event format, and event hosts. We dedicated the event to the former SJAA president Gerry Joyce who passed away earlier that week. Gerry was a leading proponent for SJAA virtual events including the Armchair star party. Then Rashi & Carl conducted a virtual tour of the night sky using Stellarium. Rashi had also prepared a fun stargazing exercise for the audience to do after the event. Next, I introduced astronomy gear focusing on telescopes. Then came the Armchair Stargazing with live views. Glenn, Bruce, Paolo, Francesco from the imaging team orchestrated a great show consisting of live views, stock pictures, and Stellarium walk-throughs. Objects included had M3 Globular cluster, M44 Beehive cluster, M97 Owl Nebula, M101 Pinwheel Galaxy, and NGC2244 Rosette Nebula. Last part of the event was a Q&A session using

YouTube questions moderated by Sukhada & Emi. We concluded the event after nearly two hours. Meetup had over 100 sign-ups for the event. We hope to improve upon this first experience and offer future Armchair Star Parties to the public.



Gerry Joyce
SJAA President
Our first 'Armchair Star Party' is dedicated to you.

Party Hosts (SJAA Members)

 Kanch	 Carl	 Rashi	 Glenn	 Bruce	 Paolo
 Francesco	 Paul	 Wolf	 Sukhada	 Emi	

The Plan

1. The Night Sky
2. Astro Gear
3. Armchair Observing
4. Q & A

Designed by: Rashi Girhar (SJAA)

Southern Sky from New Zealand—Observation Report

Credit: Marilyn Perry

I have wanted to see the Southern Skies ever since I first got interested in star gazing. In 2013 I got a glimpse of how spectacular it is from a cruise ship near Aruba when I saw the Southern Cross just clearing the ocean horizon. In addition to the Southern Cross I was blown away by many bright stars in Centaurus and Carina that surrounded it. I got just a quick glimpse before the sun rose and my star gazing was over. I was left with bucket list items of a better view of the Southern Cross and vicinity and a view of the Large and Small Magellanic Clouds (LMC/SMC).

When my son married a Kiwi, I knew I had to take my family to New Zealand so I could see her beautiful homeland and hopefully I could see the Southern Skies. As our November 2019 Road Scholar tour approached, I kept checking the long range weather forecast and it did not look promising. The forecast for almost every day and night was for rain but sometimes it improved to partly cloudy. Our tour did include a night at the International Dark Sky Reserve by Aoraki/Mount Cook in the Southern Alps of the South Island where the star gazing can be spectacular; unfortunately the long range weather forecast called for only rain or snow there. I have a backyard in Sunnyvale which has lots of sky glow but is sheltered from direct street lights, I do the vast majority of my star gazing in my backyard, and I was determined to use my experience of star gazing in less than optimum conditions to see the Southern Skies the best I could.

In New Zealand the weather switched between clear skies to cloudy skies to rain with hail and even a tornado thrown in. The first clear night in Auckland my family and I visited the Stardome Observatory & Planetarium, located on a hill about 5.5 miles from Auckland Central Business District. They had a planetarium show, but I was only interested in viewing the night sky from their dark back patio with an excellent southern horizon which I needed because the lowest Southern Cross star had altitude of 4 degrees. I viewed through my binoculars and through their 10" aperture telescopes. I could see the Southern Cross and its pointer stars, Alpha and Beta Centauri, very well. An astronomer showed me 47 Tucanae (NGC104). It was beautiful, the second brightest and largest globular cluster in the sky (after Omega Centauri). I asked to see LMC/SMC and was told that they were invisible due to hazy skies due to wildfires that were currently burning in Australia. It was very disorienting to see the tail of Scorpius and the "teapot" of Sagittarius upside down. When I mentioned the "teapot" to the astronomer, he told me that the "pot" wasn't visible in the sky yet. He thought I meant Orion, because in the Southern hemisphere, Orion is upside down and Orion's belt plus stars Rigel and Saiph form the shape of a pot. While waiting for our taxi to pick us up I spotted an extremely bright group of stars just clearing a building with no other stars around to give me frame of reference. I thought it was some new Southern hemisphere cluster, but felt ridiculous when Sky Safari identified it as Pleiades (M45). It just looked so different upside down! With the sky glow from Auckland and the haze from the fires, the conditions were very similar to my Sunnyvale backyard, but I thought it was a very worthwhile evening because I had seen the Southern Cross and I was getting used to the unfamiliar orientation of the stars in the sky.

As expected we were rained/snowed out of star gazing at Aoraki Mount Cook, but the Sir Edmund Hillary Alpine Centre Digital Dome planetarium was by far the best planetarium I have ever seen. The presentation was top-notch with the highlights being how to use Alpha and Beta Centauri stars as pointers to the Southern Cross so as not to get confused with two other crosses in the vicinity, "False Cross" (southern-most star is Epsilon Carinae) and "Diamond Cross" (southern-most star is Beta Carinae). I give the astronomer credit for giving us every chance to see the night sky. He didn't announce that we would have a planetarium show instead of viewing the sky until he poked his head out at the last minute and saw it was still raining. Then he gave us one last chance to view the sky by poking his head out again at the end of the planetarium show – alas, it was still raining

My last chances to view LMC/SMC were our last nights at Queenstown. Our hotel room had perfect exposure, looking south over Lake Wakatipu with the view going almost down to the horizon in only one location, right where the Southern Cross was visible. We did have the light pollution of the city and a clear view of a street light, but my wonderful niece was kind enough to position herself between me and the street light, and I could see that little part of the sky very well. Because of the lights, I needed my binoculars to see LMC/SMC and I needed to star hop to get to them, but they were clearly visible. The star hop was easy enough that my niece was able to follow my route and view them herself. They were extended smudges, not the beautiful photos shown in Sky Safari, but smudges unlike any bright nebula I had ever seen before. Highlights in the vicinity of Southern Cross were Jewel Box Cluster, (NGC4755), Lambda Centauri Cluster (IC2944), Southern Pleiades (IC2602) and Eta Carinae Nebula (NGC3372). With persistence and the knowledge that one doesn't need pristine skies in order to see objects in the sky, I was able to check off my bucket list items, good view of Southern Cross and its vicinity and LMC/SMC.

NASA Contributes Expertise, Ingenuity to COVID-19 Fight

Credits: NASA



NASA engineer Mike Buttigieg works on the Aerospace Valley Positive Pressure Helmet, a device that was successfully tested by doctors at Antelope Valley Hospital in California. The Spaceship

Company began producing 500 this week and a request was submitted April 22 to the FDA for an emergency use authorization. NASA's Armstrong Flight Research Center in California partnered with Antelope Valley Hospital, the City of Lancaster, Virgin Galactic, The Spaceship Company (TSC), Antelope Valley College and members of the Antelope Valley Task Force to solve possible shortages of critical medical equipment in the local community.

Credit: NASA

NASA has joined the fight against coronavirus (COVID-19) with efforts underway across the country to augment the national response, a few of which were highlighted in a media briefing today.

"NASA's strength has always been our ability and passion – collective and individual – for solving problems," said NASA Administrator Jim Bridenstine. "All the work being done shows how NASA is uniquely equipped to aid in the federal response to coronavirus by leveraging the ingenuity of our workforce, mobilizing investments made in the U.S. space agency to combat this disease, and working with public and private partnerships to maximize results."

On April 1, NASA launched an agencywide call for ideas on its internal crowdsourcing platform NASA@WORK for how the agency can leverage its expertise and capabilities to help the nation with this unprecedented crisis. In just two weeks, 250 ideas were submitted, more than 500 comments were submitted, and more than 4,500 votes were cast.

In addition to the NASA@Work challenge, the agency workforce developed ideas and worked with partners to quickly respond to the health crisis within the last month. Agency efforts highlighted during the media briefing include:

VITAL Ventilator

Engineers at NASA's Jet Propulsion Laboratory in California designed a new high-pressure ventilator tailored specifically to treat COVID-19 patients. The device, called VITAL (Ventilator Intervention Technology Accessible Locally), passed a critical test on April 21 at the Icahn School of Medicine at Mount Sinai in New York – an epicenter of COVID-19 in the United States – and now is under review for an emergency use authorization by the Food and Drug Administration (FDA).

VITAL is designed to treat patients with milder symptoms, thereby keeping the nation's limited supply of traditional ventilators available for patients with more severe COVID-19 symptoms.

The device can be built faster and maintained more easily than a traditional ventilator, and is comprised of far fewer parts, making it more economical to produce. It was designed to use parts currently available to potential manufacturers but not compete with the existing supply chain of currently made ventilators.

Aerospace Valley Positive Pressure Helmet

NASA's Armstrong Flight Research Center in California partnered with Antelope Valley Hospital, the City of Lancaster, Virgin Galactic, The Spaceship Company (TSC), Antelope Valley College and members of the Antelope Valley Task Force to solve possible shortages of critical medical equipment in the local community.

One of the task force's first efforts was to build an oxygen helmet to treat COVID-19 patients exhibiting minor symptoms and minimize the need for those patients to use ventilators. The device functions like a continuous positive airway pressure (CPAP) machine to force oxygen into a patient's low-functioning lungs.

Called the Aerospace Valley Positive Pressure Helmet, the device was successfully tested by doctors at Antelope Valley Hospital. The Spaceship Company began producing 500 this week and a request was submitted April 22 to the FDA for an emergency use authorization.

Hubble Marks 30 Years in Space With Tapestry of Blazing Starbirth

Credit: Nasa

NASA is celebrating the Hubble Space Telescope's 30 years of unlocking the beauty and mystery of space by unveiling a stunning new portrait of a firestorm of starbirth in a neighboring galaxy.

In this Hubble portrait, the giant red nebula (NGC 2014) and its smaller blue neighbor (NGC 2020) are part of a vast star-forming region in the Large Magellanic Cloud, a satellite galaxy of the Milky Way, located 163,000 light-years away. The image is nicknamed the "Cosmic Reef," because it resembles an undersea world.

A colorful image resembling a cosmic version of an undersea world teeming with stars is being released to commemorate the Hubble Space Telescope's 30 years of viewing the wonders of space. In the Hubble portrait, the giant red nebula (NGC 2014) and its smaller blue neighbor (NGC 2020) are part of a vast star-forming region in the Large Magellanic Cloud, a satellite galaxy of the Milky Way, located 163,000 light-years away. The image is nicknamed the "Cosmic Reef," because NGC 2014 resembles part of a coral reef floating in a vast sea of stars. Some of the stars in NGC 2014 are monsters. The nebula's sparkling centerpiece is a grouping of bright, hefty stars, each 10 to 20 times more massive than our Sun. The seemingly isolated blue nebula at lower left (NGC 2020) has been created by a solitary mammoth star 200,000 times brighter than our Sun. The blue gas was ejected by the star through a series of eruptive events during which it lost part of its outer envelope of material.



Some of the stars in NGC 2014 are monsters. The nebula's sparkling centerpiece is a grouping of bright, hefty stars, each 10 to 20 times more massive than our Sun. The seemingly isolated blue nebula at lower left (NGC 2020) has been created by a solitary mammoth star 200,000 times brighter than our Sun. The blue gas was ejected by the star through a series of eruptive events during which it lost part of its outer envelope of material.

Credits: NASA, ESA and STScI

Thirty years ago, on April 24, 1990, Hubble was carried aloft from NASA's Kennedy Space Center in Florida aboard the space shuttle Discovery, along with a five-astronaut crew. Deployed into Earth orbit a day later, the telescope opened a new eye onto the cosmos that has been transformative for our civilization.

Hubble is revolutionizing modern astronomy, not only for scientists, but also by taking the public on a wondrous journey of exploration and discovery. Hubble's never-ending, breathtaking celestial snapshots provide a visual shorthand for Hubble's top scientific achievements. Unlike any space telescope before it, Hubble made astronomy relevant, engaging and accessible for people of all ages. The space telescope's iconic imagery has redefined our view of the universe and our place in time and space.

"Hubble has given us stunning insights about the universe, from nearby planets to the farthest galaxies we have seen

so far," said Thomas Zurbuchen, associate administrator for science at NASA Headquarters in Washington, D.C. "It was revolutionary to launch such a large telescope 30 years ago, and this astronomy powerhouse is still delivering revolutionary science today. Its spectacular images have captured the imagination for decades, and will continue to inspire humanity for years to come."

Unencumbered by Earth's blurring atmosphere, the space observatory unveils the universe in unprecedented crystal-clear sharpness across a broad range of wavelengths, from ultraviolet to near-infrared light.

Hubble's top accomplishments include measuring the expansion and acceleration rate of the universe; finding that black holes are common among galaxies; characterizing the atmospheres of planets around other stars; monitoring weather changes on planets across our solar system; and looking back in time across 97% of the universe to chronicle the birth and evolution of stars and galaxies.

Hubble has yielded to date 1.4 million observations and provided data that astronomers around the world have used to write more than 17,000 peer-reviewed scientific publications, making it the most prolific space observatory in history. Its archival data alone will fuel future astronomy research for generations to come.

Hubble's longevity can be attributed to five space shuttle servicing missions, from 1993 to 2009, in which astronauts upgraded the telescope with advanced instruments, new electronics and on-orbit repairs. The venerable observatory, with its suite of cameras and other instruments, is expected to stay operational through the 2020s, in synergy with the upcoming James Webb Space Telescope.

The Hubble Space Telescope is a project of international cooperation between NASA and ESA (the European Space Agency). NASA's Goddard Space Flight Center in Greenbelt, Maryland, manages the telescope. The Space Telescope Science Institute (STScI) in Baltimore conducts Hubble science operations. STScI is operated for NASA by the Association of Universities for Research in Astronomy in Washington, D.C.



Date: April 19, 2019

Location: Sierra Nevada, Yosemite

Imaging Telescope: Celestron RASA

Imaging Camera: QHY168C

Mount: Software Bisque Paramount MX+

Software: StarTools , TheSkyX Pro , SkyTools , SharpCap 3.1 Sharp Cap , PixInsight1.8 , PHD Guiding 2 PHD 2 Guiding



Open Cluster NGC 2547 at Vela
Nikola Nikolov

Dates: April 1 2020, April 4 2020, April 5 2020

Location: Siding Springs Observatory, Coonabarabran, New South Wales, Australia

Telescope: Takahashi FSQ ED 106

Camera: FLI MicroLine 11002 Grade 1 Monochrome CCD

Mount: Paramount PME

Software: Photoshop CC 2018 , AstroPixelProcessor , Adobe Lightroom CC

SJAA Member Astrophoto Gallery

Pinwheel Galaxy
Jian Yuan Peng



Date: March 30, 2020

Location: Courtyard, Palo Alto, CA

Telescope: TEC 160FL

Camera: ZWO ASI1600MM-Cool Pro

Mount: IOptron CEM60.

Software: SePHD2, PixInsight, Lightroom, Sequence Generator Pro, Photoshop Sequence Generator Pro, Photoshop

SJAA Member Astrophoto Gallery



Dates: Sept. 1, 2019 , Dec. 27, 2019 , Jan. 9, 2020 , Feb. 14, 2020

Location: San Jose, CA

Telescope: Stellarvue SV4

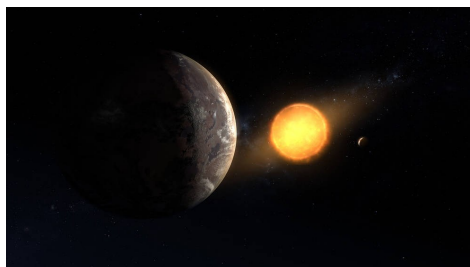
Camera: ZWO ASI1600MM-Cool

Mount: Astro-Physics Mach1 GTO

Software: Photoshop , PHD2 , Sequence Generator Pro , PixInsight

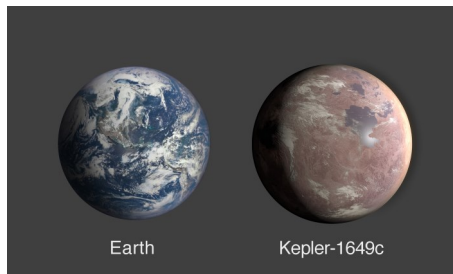
Earth-Size, Habitable Zone Planet Found Hidden in Early NASA Kepler Data

Credit NASA



An illustration of Kepler-1649c orbiting around its host red dwarf star. This newly discovered exoplanet is in its star's habitable zone and is the

closest to Earth in size and temperature found yet in Kepler's data.



A comparison of Earth and Kepler-1649c, an exoplanet only 1.06 times Earth's radius

Credits: NASA/Ames Research Center/Daniel Rutter

A team of transatlantic scientists, using reanalyzed data from NASA's Kepler space telescope, has discovered an Earth-size exoplanet orbiting in its star's habitable zone, the area around a star where a rocky planet could support liquid water.

Scientists discovered this planet, called Kepler-1649c, when looking through old observations from Kepler, which the agency retired in 2018. While previous searches with a computer algorithm misidentified it, researchers reviewing Kepler data took a second look at the signature and recognized it as a planet. Out of all the exoplanets found by Kepler, this distant world – located 300 light-years from Earth – is most similar to Earth in size and estimated temperature.

This newly revealed world is only 1.06 times larger than our own planet. Also, the amount of starlight it receives from its host star is 75% of the amount of light Earth receives from our Sun – meaning the exoplanet's temperature may be similar to our planet's, as well. But unlike Earth, it orbits a red dwarf. Though none have been observed in this system, this type of star is known for stellar flare-ups that may make a planet's environment challenging for any potential life.

"This intriguing, distant world gives us even greater hope that a second Earth lies among the stars, waiting to be found," said Thomas Zurbuchen, associate administrator of NASA's Science Mission Directorate in Washington. "The data gathered by missions like Kepler and our Transiting Exoplanet Survey Satellite (TESS) will continue to yield amazing discoveries as the science community refines its abilities to look for promising planets year after year."

Credits: NASA/Ames Research Center/Daniel Rutter

There is still much that is unknown about Kepler-1649c, including its atmosphere, which could affect the planet's temperature. Current calculations of the planet's size have significant margins of error, as do all values in astronomy when studying objects so far away. Rocky planets orbiting red dwarfs are of particular astrobiological interest. However, astrobiologists will need much more information about this planet in order to gauge whether it is promising for life as we know it. But based on what is known, Kepler-1649c is especially intriguing for scientists looking for worlds with potentially habitable conditions.

There are other exoplanets estimated to be closer to Earth in size, such as TRAPPIST-1f and, by some calculations, Teegarden c. Others may be closer to Earth in temperature, such as TRAPPIST-1d and TOI 700d. But there is no other exoplanet that is considered to be closer to Earth in both of these values that also lies in the habitable zone of its system.

"Out of all the mislabeled planets we've recovered, this one's particularly exciting – not just because it's in the habitable zone and Earth-size, but because of how it might interact with this neighboring planet," said Andrew Vanderburg, a researcher at the University of Texas at Austin and first author on the paper released today in *The Astrophysical Journal Letters*. "If we hadn't looked over the algorithm's work by hand, we would have missed it."

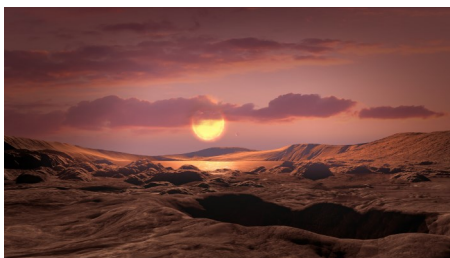
Kepler-1649c orbits its small red dwarf star so closely that a year on Kepler-1649c is equivalent to only 19.5 Earth days. The system has another rocky planet of about the same size, but it orbits the star at about half the distance of Kepler-1649c, similar to how Venus orbits our Sun at about half the distance that Earth does. Red dwarf stars are among the most common in the galaxy, meaning planets like this one could be more common than we previously thought.

Looking for False Positives

Previously, scientists on the Kepler mission developed an algorithm called Robovetter to help sort through the massive amounts of data produced by the Kepler spacecraft, managed by NASA's Ames Research Center in California's Silicon Valley. Kepler searched for planets using the transit method, staring at stars, looking for dips in brightness as planets passed in front of their host stars.

Most of the time, those dips come from phenomena other than planets – ranging from natural changes in a star's brightness to other cosmic objects passing by – making it look like a planet is there when it's not. Robovetter's job was to distinguish the 12% of dips that were real planets. Those signatures Robovetter determined to be from other sources were labeled "false positives," the term for a test result mistakenly classified as positive.

With an enormous number of tricky signals, astronomers knew the algorithm would make mistakes and would need to be double-checked – a perfect job for the Kepler False Positive Working Group. That team reviews Robovetter's work, going through all false positives to ensure they are truly errors and not exoplanets, ensuring fewer potential discoveries are overlooked. As it turns out, Robovetter had mislabeled Kepler-1649c



An illustration of what Kepler-1649c could look like from its surface.

Credits: NASA/Ames Research Center/Daniel Rutter

Even as scientists work to further automate analysis processes to get the most science as possible out of any given dataset, this discovery shows the value of double-checking automated work. Even six years after Kepler stopped collecting data from the original Kepler field – a

patch of sky it stared at from 2009 to 2013, before going on to study many more regions – this rigorous analysis uncovered one of the most unique Earth-analogs discovered yet.

A Possible Third Planet

Kepler-1649c not only is one of the best matches to Earth in terms of size and energy received from its star, but it provides an entirely new look at its home system. For every four times the outer planet in the system orbits the host star, the inner planet orbits almost exactly nine times. The fact that their orbits match up in such a stable ratio indicates the system itself is extremely stable, and likely to survive for a long time.

Nearly perfect period ratios are often caused by a phenomenon called orbital resonance, but a nine-to-four ratio is relatively unique among planetary systems. Usually resonances take the form of ratios such as two-to-one or three-to-two. Though unconfirmed, the rarity of this ratio could hint to the presence of a middle planet with which both the inner and outer planets revolve in synchronicity, creating a pair of three-to-two resonances.

The team looked for evidence of such a mystery third planet, with no results. However, that could be because the planet is too small to see or at an orbital tilt that makes it impossible to find using Kepler's transit method.

Either way, this system provides yet another example of an Earth-size planet in the habitable zone of a red dwarf star. These small and dim stars require planets to orbit extremely close to be within that zone – not too warm and not too cold – for life as we know it to potentially exist. Though this single example is only one among many, there is increasing evidence that such planets are common around red dwarfs.

"The more data we get, the more signs we see pointing to the notion that potentially habitable and Earth-size exoplanets are common around these kinds of stars," said Vanderburg. "With red dwarfs almost everywhere around our galaxy, and these small, potentially habitable and rocky planets around them, the chance one of them isn't too different than our Earth looks a bit brighter."

Missions such as Kepler and TESS help contribute to the field of astrobiology, the interdisciplinary research into understanding how the variables and environmental conditions of distant worlds could harbor life as we know it, or whatever other form that life could take.



Kid Spot Jokes:

- ♦ **How far can you see on a clear day ?**
150 million kilometers, from here to the Sun.
- ♦ **When is the moon the heaviest?**
When it's full
- ♦ **What do you think of that new restaurant on the moon ?**
The food's great but it has no atmosphere.

Fun Facts

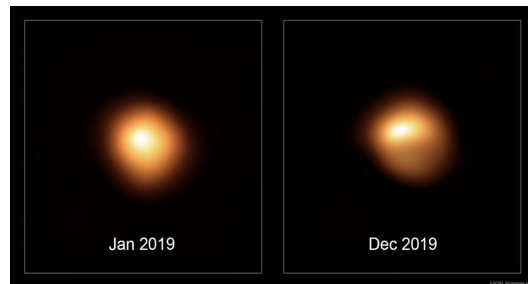
Neutron Stars



Credit: X-ray (NASA/CXC/
ESO/F. Vogt et al.)

- What are they? Neutron stars are nothing but very highly compressed remnants of a massive star that died in a supernova event
- The most massive neutron star ever detected, was just last year by the Green Bank Observatory. It is called J0740+6620, is about 4600 light years from Earth and 2.14 times the mass of the sun!
- Typical diameters of neutron stars are about 20-40km in diameter with about twice the mass of our Sun being compressed in a much much smaller volume. (Just for comparison, our Sun is about 1.39 million kilometers in diameter, about 35000 times that of these average neutron stars!)
- Another interesting comparison would be that the surface temperature of neutron stars could be as high as 60,000K compared to 6,000K for the Sun.

Betelgeuse



The Changing Surface of Fading Betelgeuse
Credit : ESO, M. Montargès et al.

- Betelgeuse is a red supergiant star that stirred quite the debate in November 2019 - January 2020 when it lost close to 40% of its brightness when compared to that in January 2019! The most probable cause of this phenomena is thought to be that Betelgeuse belched up a big cloud of dust and gas. This dust expanded in our line of sight to Betelgeuse that caused the apparent dimness. It's now almost back to its original brightness.
- Betelgeuse is one of the stars in Orion the Hunter, and is about a 1000 times larger than our Sun, but is only about 10 million years old compared to our sun which is about 4.5 billion years old!
- The diameter of Betelgeuse is larger than the orbital diameter of Mars!
- It is interesting that the surface temperature of Betelgeuse is only 3200C (60% as hot as the Sun), but is 100,000 times more luminous than the Sun due to its enormous size!

From the Board of Directors

Announcements

- ◇ San Jose Astronomical Association is closely monitoring the coronavirus/COVID-19 situation in Santa Clara County. In accordance with guidance from Santa Clara County Health Department, SJAA leaders have been canceling many events, especially those that involve close contact and shared surfaces on telescopes. Some events, however, have successfully been moved to an online format. Please check our Meetup to determine if an event has been canceled or moved online. The board of directors will periodically review the status of the outbreak and the guidance from the county to determine the impact upon upcoming events.

- ◇ Sadly on the 20th of May the SJAA lost its President Gerry Joyce. For information regarding his obituary please see Page 2 of this issue.

- ◇ Due to the unexpected passing of SJAA's President Gerry Joyce, Vice-President Kenichi Miura will assume the role of Acting President until such time that the board chooses a new President.

Board Meeting Excerpts

March 7, 2020

In attendance

Swami, Sandy, Glenn, Ken, Gerry, Peter Wolf, Sukhada, Rob J
Excused Absences: Vini
Guests: Marianne, Emi

Officer Elections

Officer elections for President, Vice-President, Treasurer, and Secretary were held during the March board meeting. Rob Jaworski was re-elected as Treasurer. Gerry Joyce was elected as President replacing Swami Nigam after his 2 terms passed. Ken Miura was elected as Vice-President replacing Gerry Joyce. Emi Nikolov was elected as Secretary replacing

Sandy Mohan who will still serve as a board member.

April 11, 2020

In attendance

Swami, Sandy, Vini, Glenn, Sukhada, Wolf, Vini, Rob J, Ken, Gerry, Peter, Emi

Excused Absences: None

Unexcused Absences: Rob J

Guests: Tom H, Vijay K, Marianne, Kanch, Gary

Virtual Meetings

Some of the programs are temporarily switching to online meetings and presentations due to COVID-19.

Stars Over Yosemite

Stars Over Yosemite star party is scheduled for July 10-11. Ken Miura is coordinating the event. Future of this event for 2020 is unknown due to the current closure of Yosemite because of Covid-19 pandemic.

May 9, 2020

In attendance

Swami, Wolf, Vini, Rob J, Glenn, Emi, Ken, Swami, Peter

Excused Absences: None

Unexcused Absences: Sandy

Guests: Marianne, Gary

Full Moon Rise Event

OSA Facebook live streaming event of the full moon rise on Thursday, May 7 from Coyote Valley. SJAA volunteers Gerry, Francesco, Joe Lin, Swami joined the FB live event and answered questions from the public through comments.

Programs that are available online

Here is the list that are of the programs that are currently available due to the COVID-19 pandemic. The programs are: Board Meeting, SIG, and ITSP and Starry Nights Star Party combined together. For a summary of the Armchair Star Party please visit Page 3 of this issue.

New Open Positions

New positions to recruit for: SJAA Liaison, Volunteer Chair

June 6, 2020

In attendance

Ken, Emi, Rob J, Sandy, Wolf, Vini, Peter, Sukhada

Excused Absences: None

Guests: Rashi, Marianne, Gary, Bruce F

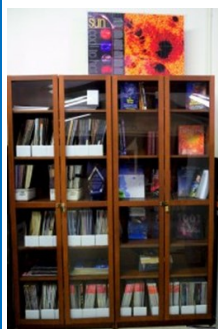
Ken Miura to continue as Acting President

Ken will continue as the Acting President until the board chooses a new President.

Stars Over Yosemite Update

Stars Over Yosemite which is scheduled for July 10th and 11th is pending due to the status of Covid-19. Any further development will be posted on the SJAA website.

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick START program is undergoing a redesign to make it more accessible to SJAA members, so signups are currently on hold.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night

sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for

additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President:	Vacant
Vice-President/Acting	
President/Director:	Ken Miura
Treasurer/Dir:	Rob Jaworski
Secretary:	Emi Nikolov
Director:	Vini Carter
Director:	Sandy Mohan
Director:	Wolf Witt
Director:	Swami Nigam
Director:	Peter Melhus
Director:	Glenn Newell
Director:	Sukhada Palav
Ephemeris Newsletter -	
Content Editor:	Abishek Ramasubramanian
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
In-Town Star Party:	Muditha Kanchana
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Rob Jaworski
Quick START	Dave Ittner
Donations:	Dave Ittner*
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Media Manager:	Jessica Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
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San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

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Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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